Alexandria N. Hughes, PhD

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Summary

Mathematical Statistician at the National Science Foundation, National Center for Science & Engineering Statistics (NSF-NCSES). Trained as both a statistician & neuroscientist with expertise in biological and health data science.

Education

Aug 2015 - Sep 2020 | **University of Colorado, Anschutz Medical Campus**
PhD, Neuroscience

Jan 2021 - Aug 2023 | **Texas A&M University**
MS, Statistics

Aug 2010 - Dec 2014 | **Boise State University**
BS, Biology; Mathematics minor

Work Experience

Dec 2022 - present | **National Science Foundation (NSF/NCSES)**
Mathematical Statistician

Oct 2021 - Nov 2022 | **Westat, Public Health & Epidemiology Practice**
Senior Biostatistician

- Designed and performed statistical analyses in R in support of scientific manuscripts and other projects; wrote statistical analysis plans, managed data dictionaries, cloud computing (AWS)

Jun 2021 - Oct 2021 | **National Institutes of Health (NIH/NICHD)**
IRTA Postdoctoral Fellow

- Analysis of scRNA-seq data with R (Bioconductor), with focus on graph-based clustering & batch correction methods

Aug 2015 - May 2021 | **University of Colorado, School of Medicine**
National Science Foundation Graduate Research Fellow; Postdoc (2020-2021)

- Molecular genetics, microscopy, and image analysis to understand neurodevelopment using zebrafish
- Recipient of National Science Foundation award (NSF-GRFP) ($138,000) to do independent research
- Published 3 primary author papers & gave 8 invited talks on projects for which I performed all experimental design, experimentation, & statistical analysis

May 2011 - Jul 2015 | **Boise State University, Dept of Mathematics**
Adjunct faculty (2014-2015), drop-in center tutor (2011-2014), TA (2011-2013)

- Taught 2 sections of Elementary Algebra, TA'ed 2 sections of Calculus II and 3 sections of Calculus III, and provided drop-in tutoring services for the dept

Skills

Programming	R, Julia, SAS, Python
Markup	LaTeX, Markdown (RMarkdown, Distill for website building), basic HTML/CSS
Bioinformatics	Bioconductor, RNA- and scRNA-seq analysis (scater, scran, edgeR, DEseq2) and Seurat, basic preprocessing with UNIX, UCSC Genome Browser, NCBI, Ensembl, Image analysis (ImageJ)
Visualization	Data visualization (ggplot2, matplotlib, Shiny), vector graphics & scientific illustration (Inkscape)
Other	RStudio, Git, Jupyter, writing, teaching & presenting, communicating mathematical/statistical concepts to non-math audiences, project management and strategy

Membership & Certification

- Member, American Statistical Association
- SAS 9.4 Certified Associate

Achievements

- University of Colorado School of Medicine Dean's Outstanding Dissertation Award, 2021: Dean's selected dissertation of the 2020-2021 doctoral graduating classes
- Invited speaker at 3 high-profile conferences in neuroscience: Cold Spring Harbor Laboratory annual meetings on Glia in Health & Disease (2018) and Zebrafish Circuits & Behavior (2019) and the Gordon Research Seminar on Glial Biology (2019), as well as additional regional conferences in Colorado
- Invited seminar series speaker at New York University (2019) and Regis University (2019)
- National Science Foundation Graduate Research Fellowship Program (NSF-GRFP) awardee, 2016: funded my tuition and stipend for 3 years of my PhD

Publications

- Andreev A, Komatsu V, Almiron P, Rose K, **Hughes AN**, Lee MY (2022). Research Culture: Welcome to the lab. *eLife*, 11:e79627.
- **Hughes AN** (2022). In search of a backup career plan, I found a perfect fit. *Science* (New York, NY), 376(6589), 210-210.
- **Hughes AN** (2021). Glial cells promote myelin formation and elimination. *Frontiers in Cell and Developmental Biology*, 9, 661486.
- **Hughes AN** & Appel B (2020). Microglia phagocytose myelin sheaths to modify developmental myelination. *Nature Neuroscience*, 23, 1055–1066.
- **Hughes AN** & Appel B (2019). Oligodendrocytes express synaptic proteins that modulate myelin sheath formation. *Nature Communications*, 10(1), 1-15.
- **Hughes AN**, Oxford A, Tawara K, Jorcyk C, Oxford JT (2017). Endoplasmic reticulum stress and unfolded protein response in cartilage pathophysiology; contributing factors to apoptosis and osteoarthritis. *International Journal of Molecular Sciences*, 18(3), 665.
- **Hughes AN**, Hixon AM, Josey M (2016). A Schwanncentric View of Axon Arborization in Neuromuscular Junction (NMJ) Formation. *Journal of Neuroscience*, 36(38), 9760-9762.
- **Hughes AN** & Oxford JT (2014). A lipid-rich gestational diet predisposes offspring to nonalcoholic fatty liver disease: a potential sequence of events. *Hepatic Medicine*, 6, 15-23.